

# Saad Ahmed Qadeer

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## HONORS & INTERNATIONAL COMPETITIONS

- Earned a **Bronze Medal** for outstanding academic performance in undergraduate studies.
- Ranked **42nd globally** out of 13,779 participants in the [Meta Hacker Cup 2025 \(AI Track\)](#)
- [Harvard University CS50x](#) Puzzle Day Achiever
- Completed [Millennium Fellowship](#), a United Nations Academic Impact leadership program.
- Participated in **10+ international competitions** including Stanford, Google DeepMind, Kaggle, etc.
- Achieved **Top-Rated** status as a Freelance Data Science Consultant, securing a 100% Job Success Score.

## EDUCATION

### Bachelor of Science in Data Science

February 2021 – January 2025

*The Islamia University of Bahawalpur, Pakistan* CGPA: 3.5 / 4.0 **Bronze Medalist**

- **Senior Year Project:** “Automated Thoracic Disease Detection via Multilabel Deep Learning Ensembles”, Achieved 0.780 AUC using a Stacking Ensemble with Grad-CAM interpretability.

## RESEARCH & ACADEMIC PROJECTS

### Multilabel Chest X-Ray Pathology Classification 🔒

*TensorFlow · Keras · DenseNet-121*

- Built an end-to-end multilabel pipeline for 7,376 NIH ChestX-ray14 radiographs.
- Fine-tuned a DenseNet-121 and ResNet50V2 stacking ensemble using Asymmetric Loss.
- Achieved a **0.780 macro AUC-ROC (0.862 for Effusion)** for class imbalance.
- Validated model attention maps using GradCAM to maximize independent F1 scores.

### Mechanistic Interpretability & In-Context Learning 🔒

*GPT-2-Small · TensorFlow · Python*

- Replicated Wang et al. (EMNLP '23) on GPT-2-Small, achieving 80.5% probing accuracy
- Formalized the **Label Surface Form Sensitivity (LSFS) metric** to expose gaps in Anchor Theory.
- Revealed an 11–12% accuracy collapse across semantically equivalent labels.
- Engineered **Canonical Anchor Projection**, a training-free embedding correction.
- Recovered 80–100% of the accuracy gap while maintaining the canonical baseline.

### De Novo RNA 3D Structure Prediction (Stanford AI Research) 🔒

*PyTorch · RhoFold+*

- Deployed a RhoFold+ pipeline, utilizing MSA subsampling to generate diverse structural ensembles.
- Designed a physics-informed random walk fallback for ultra-long sequences (>440 nt).
- Bypassed embedding limits and enforced valid C1' backbone geometry.
- Achieved a peak TM-score of **0.507** (correct global fold topology) on the 9IWF target.
- Set a **0.139 zero-shot baseline** without task-specific fine-tuning.

### NYC Taxi Revenue Optimization & Predictive Modeling 🔒

*Python · scikit-learn · XGBoost · statsmodels*

- Built an end-to-end pipeline from data analysis to predictive modeling on taxi trip data.
- Used hypothesis testing to confirm payment type significantly affects fare spend.
- Modeled fares with linear regression, finding trip distance the strongest predictor.
- Trained an XGBoost classifier to flag high-tip riders at **AUC > 0.85**, outperforming Random Forest.

### Probabilistic RUL Estimation for Turbofan Engines 🔒

*PyTorch · MLflow · Numpy*

- Engineered a Bayesian LSTM with MC Dropout to approximate deep Gaussian posteriors.
- Achieved a **competitive 15.18 RMSE** on the NASA C-MAPSS FD001 benchmark dataset.
- Prevented data leakage via train-only scaling and sliding-window tensor construction.
- Quantified epistemic uncertainty with 95% prediction intervals that narrow near failure.
- Bounded catastrophic late-prediction risks utilizing the **NASA Asymmetric Score**.

## WORK EXPERIENCE

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### Statistical Data Analyst

February 2025 – October 2025

N5 Sensors

remote, USA

- Collaborated with PhD researchers to optimize detection algorithm thresholds.
- Maximized true-positive rates and suppressed false alarms in early fire warning systems.
- Developed anomaly-detection pipelines and conducted LTE connectivity analysis (RSRP, RSRQ).
- Performed end-to-end statistical analysis of real-world fire events and gas detection data.

### Independent Data Science Consultant

July 2022 – Present

Upwork (International Clients)

Remote

- Maintained a **100% Job Success Score** across 6+ projects in predictive modelling and data science.

## KEY SKILLS AND TECHNOLOGIES

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**Programming Languages:** Python, SQL, C++, MATLAB, LaTeX

**Machine & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, OpenCV, XGBoost

**Artificial Intelligence:** Hugging Face, LangChain, LLM fine tuning, AI Agents

**Data Science:** Pandas, NumPy, SciPy, Statsmodels, Matplotlib, Seaborn, Power BI

**Development & Tools:** Git, Docker, Linux, Jupyter Notebook, VS Code, AWS

## TEACHING EXPERIENCE

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### Instructor – “Advanced Statistics with Projects”

February 2026 – March 2026

iCodeGuru, California, USA

- Design and taught an “Advanced Statistics with Projects” course to 40+ underprivileged students.
- Guided students through practical data science applications and hands-on statistical modeling.
- Strengthened public speaking and technical communication through daily presentations.
- All course sessions are fully documented and recorded, accessible via [Sheets](#).

### Instructor – “Introduction to Generative AI with LangChain and Hugging Face”

May 2026

iCodeGuru, California, USA

- Delivered workshop title “Introduction to Generative AI with LangChain and Hugging Face”.
- Covered concepts of generative AI, LangChain, and Hugging Face through practical demonstrations.
- Recorded sessions are available at [Workshop Recording](#).

## HOBBIES

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- Morning walks, cooking, and recreational sports (cricket and badminton).